

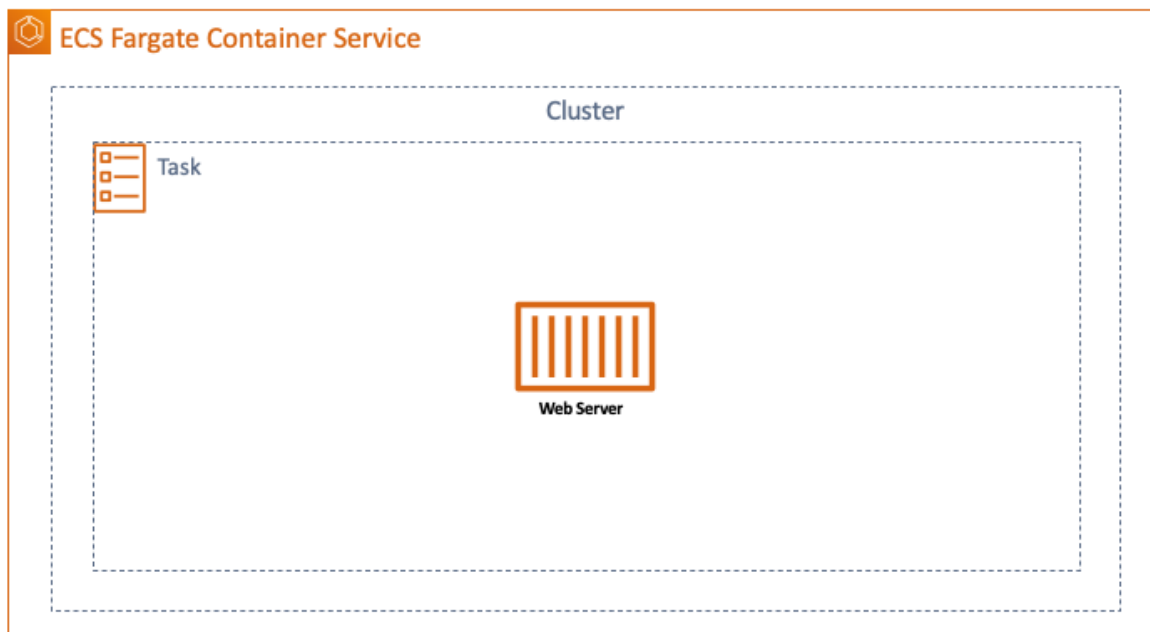
Lab-020

Running an Amazon ECS Sample App

Goal

This lab illustrates how to launch a web server using Amazon ECS's [Fargate service](#).

Architecture Diagram



Overview

Fargate is a fully managed container service that automatically allocates computing resources to run containers with scaling capabilities. To best understand Amazon's Elastic Container Service let's break it into components:

- Container: packaging of an application including code, runtime, system tools, libraries, and everything else needed to run an application (a container is created from an image);
- Cluster: a logical group of tasks or services (this lab will create an ECS cluster containing a single task);
- Task: specified in JSON format, a task defines what to do (the application itself), where to run it (minimum infrastructure requirements), and how to run it (logging, scaling configurations, security configurations, etc.);
- Service: number of instances of tasks to run and launching parameters.

Note that a cluster can contain tasks that are running on distinct containers.

To start this lab go to ECS and click on *Get started*. This lab will create a Fargate cluster using the *sample-app* template.

Step 1 - Choose a Container's Image

Select the *sample-app* container image which has definitions to run a web server using a single task.

Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate

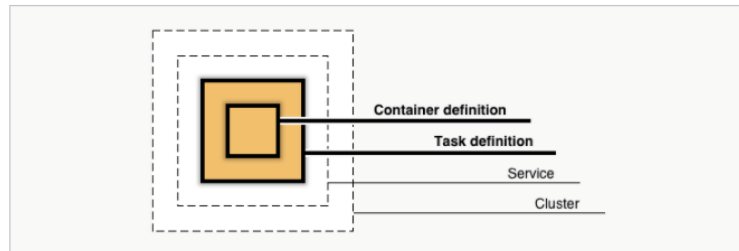
Step 1: Container and Task

Step 2: Service

Step 3: Cluster

Step 4: Review

Diagram of ECS objects and how they relate



Container definition

Edit

Choose an image for your container below to get started quickly or define the container image to use.

sample-app image : httpd:2.4 memory : 0.5GB (512) cpu : 0.25 vCPU (256)	nginx image : nginx:latest memory : 0.5GB (512) cpu : 0.25 vCPU (256)
tomcat-webserver image : tomcat memory : 2GB (2048) cpu : 1 vCPU (1024)	custom image : -- memory : -- cpu : -- Configure

Task definition

Edit

A task definition is a blueprint for your application, and describes one or more containers through attributes. Some attributes are configured at the task level but the majority of attributes are configured per container.

Task definition name	first-run-task-definition	?
Network mode	awsvpc	?
Task execution role	Create new	?
Compatibilities	FARGATE	?
Task memory	0.5GB (512)	
Task CPU	0.25 vCPU (256)	

*Required

Cancel

Next

Step 2 - Provide Service Settings

For lab you can accept the default settings.

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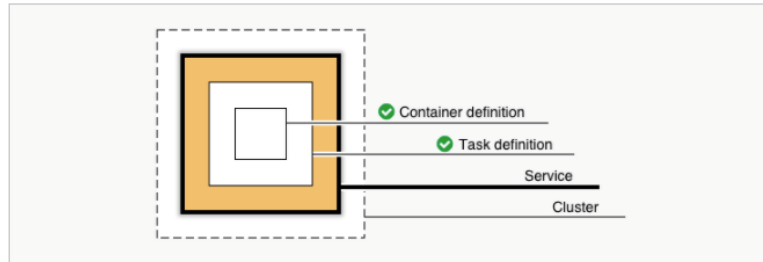
[Step 1: Container and Task](#)

Step 2: Service

[Step 3: Cluster](#)

[Step 4: Review](#)

Diagram of ECS objects and how they relate



Define your service

Edit

A service allows you to run and maintain a specified number (the "desired count") of simultaneous instances of a task definition in an ECS cluster.

Service name **sample-app-service**

Number of desired tasks **1**

Security group **Automatically create new**

A security group is created to allow all public traffic to your service only on the container port specified. You can further configure security groups and network access outside of this wizard.

Load balancer type ☒ None
☐ Application Load Balancer

*Required

[Cancel](#)

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Step 3 - Configure your Cluster

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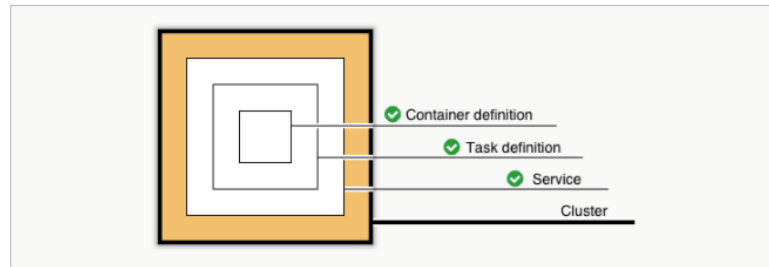
[Step 1: Container and Task](#)

[Step 2: Service](#)

Step 3: Cluster

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Diagram of ECS objects and how they relate



Configure your cluster

The infrastructure in a Fargate cluster is fully managed by AWS. Your containers run without you managing and configuring individual Amazon EC2 instances.

To see key differences between Fargate and standard ECS clusters, see the [Amazon ECS documentation](#).

Cluster name

Cluster names are unique per account per region. Up to 255 letters (uppercase and lowercase), numbers, and hyphens are allowed.

VPC ID



Subnets



*Required

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Step 4 - Review

Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate

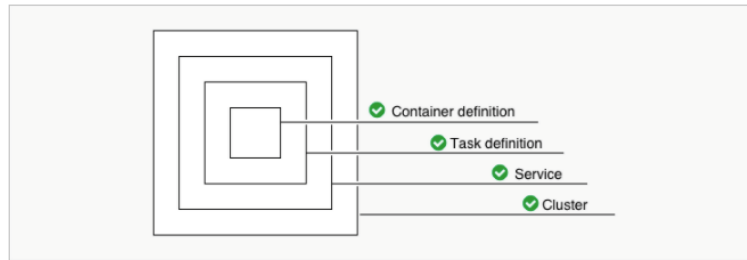
[Step 1: Container and Task](#)

[Step 2: Service](#)

[Step 3: Cluster](#)

Step 4: Review

Diagram of ECS objects and how they relate



Review

Review the configuration you've set up before creating your task definition, service, and cluster.

Task definition

Edit

Task definition name first-run-task-definition

Network mode awsvpc

Task execution role Create new

Container name sample-app

Image httpd:2.4

Memory 512

Port 80

Protocol HTTP

Service

Edit

Service name sample-app-service

Number of desired tasks 1

Cluster

Edit

Cluster name lab-020

VPC ID Automatically create new

Subnets Automatically create new

*Required

Cancel

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Create

When the configuration is deployed, click on *view service*.

Test and Validation

Click on *Tasks* and then on the single task displayed. Copy the public IP and test it using a browser. You should be able to see the web server initial page.

Amazon ECS Sample App

Congratulations!

Your application is now running on a container in Amazon ECS.